Getting Unstuck: Rehabilitation Guide After Lumbosacral Fusion

Based on research published up to October 2026. Use it together with your physiotherapist, pain doctor and surgeon, not instead of them.

Why there is reason for hope

- Long-lasting pain after spinal surgery is common and has a name: **persistent spinal pain syndrome type 2**. It is recognised, researched and treatable.
- Pain that lasts for years is driven partly by a **sensitised nervous system**: the alarm system turns the volume up. The pain is real, but it no longer reliably measures damage, and a sensitised system can be calmed down.
- The best results in recent trials come from **combining** movement, understanding pain, better sleep, psychological skills and a return to meaningful activity. No single treatment fixes everything, but several together often do.
- Improvement is possible even years after surgery. You are not too late.

1. What the newest research shows

Approach	What it is	What the evidence says
Cognitive functional therapy	Physiotherapist-led. Retrains feared and painful movements in a relaxed way, changes unhelpful beliefs about the back, and builds activity and lifestyle habits.	RESTORE trial (492 people, The Lancet 2023; three-year follow-up, The Lancet Rheumatology 2025): seven sessions plus one booster gave clearly larger and lasting gains in daily function and pain than usual care, still present after three years.
Exercise therapy	Strength, Pilates and motor control training.	Network meta-analyses (British Journal of Sports Medicine 2019; Journal of Orthopaedic and Sports Physical Therapy 2022) rank Pilates, strength and core training highest. Your two programs already cover this.
Walking	A progressive, personal walking plan.	WalkBack trial (701 people, The Lancet 2024): walking plus education cut activity-limiting relapses by about 28 percent and care-seeking by 43 percent. Free and low-risk.
Rehabilitation after fusion	Exercise combined with cognitive-behavioural coaching.	Systematic review (European Spine Journal 2022): the combination reduces disability and fear of movement more than exercise alone. A Danish trial from Aarhus found better recovery in the first six months.
Pain reprocessing therapy	Psychological therapy that teaches the brain to reinterpret pain signals as safe.	One trial (151 people, Journal of the American Medical Association Psychiatry 2022): 66 percent were pain-free or nearly pain-free after treatment versus 10 percent with usual care; over half still were after five years (reported 2026). Not yet tested specifically after fusion.
Acceptance and commitment therapy	Psychological therapy focused on living by your values despite pain.	Reviews (Journal of Pain 2024): improves function, mood and quality of life; smaller effect on pain intensity.
Cognitive behavioural therapy for insomnia	A structured sleep programme without sleeping pills.	Reliably improves sleep in people with chronic pain (2024 meta-analysis); effect on pain itself is small.
Spinal cord stimulation	An implanted device that alters pain signals.	For persistent pain after spinal surgery: moderate-quality evidence of less pain and disability at six months; about half of trial participants reach at least 50 percent pain relief. Debated, and for carefully selected people after other options.

2. Make sure nothing has been missed

Ask your surgeon and pain doctor about these possible pain sources, which are often overlooked:

- **Fusion not fully healed (non-union) or loose screws.** These can cause pain on movement. A computed tomography scan shows whether the bone has bridged.
- **Sacroiliac joint pain.** After fusion to the sacrum, studies find the sacroiliac joint is the pain source in roughly 30 to 40 percent of people with ongoing pain. Typical signs: pain below the belt line or in the buttock, worse when standing on one leg. It can be tested by examination and a diagnostic injection.

- **The level above the fusion.** The joints between the fourth and fifth lumbar vertebrae take extra load after fusion and can hurt on backward leaning.
- **Inflammatory back pain.** Back pain that started before age 40, is worst at night and in the morning, and eases with movement fits axial spondyloarthritis, an inflammatory condition most common in young men. It is treated very differently. Ask whether it has been ruled out with blood tests and magnetic resonance imaging of the sacroiliac joints.
- **Nerve pain.** Burning, electric or tingling pain responds to different treatments than joint or muscle pain.

3. Your 12-week program plan

Your daily program (1 to 3 rounds of 15 to 20 repetitions, every day) and hard program (3 rounds of 15 to 20 repetitions to muscle fatigue, every other day) are a strong base. The plan below builds into them gradually.

Pain rule during exercise: pain may rise by up to 2 points above your usual level (on a scale from 0 to 10) if it settles within an hour and is no worse the next morning. If it is worse the next morning, do less next time, not nothing.

Weeks 1 to 4: Foundation

- Daily program: 1 to 2 rounds of 15 repetitions every day.
- Hard program: twice a week, 1 to 2 rounds. Skip number 9 (back raise on ball) and number 12 (kneeling rollout on ball) until your physiotherapist approves them, because both load backward bending.
- Walking: find your baseline (what you manage on a bad day), then walk that amount twice daily.

Weeks 5 to 8: Build

- Daily program: 2 to 3 rounds. Add the progressions in the program one at a time: opposite arm lift (number 7), one foot lifted (number 12), standing hip abduction (number 9).
- Hard program: every other day, 2 to 3 rounds. On hard-program days the daily program is optional.
- Walking: add about 10 percent per week.
- Start practising backward leaning with your physiotherapist (see section 4).

Weeks 9 to 12: Strength and life

- Hard program: 3 rounds to muscle fatigue, every other day. Try the one-leg push-up progression (number 4).
- Daily program: 1 to 2 rounds on rest days.
- Add one activity you enjoy: warm-water training, cycling, swimming or a Pilates class.
- Compare with week 1: walking time, sitting time, sleep and morning pain.

Flare-up plan. Flare-ups are expected and do not mean damage.

1. Reduce, do not stop: halve the daily program, keep short walks, pause the hard program.
2. Calm the system: heat, slow breathing, comfortable positions.
3. Return over 3 to 7 days, stepping back up one level at a time.
4. Look for the trigger, often a "boom and bust" day of doing too much.
5. Contact your physiotherapist if it has not settled within two weeks.

Track weekly: morning pain (0 to 10), walking minutes, hours slept, one thing you did that mattered to you.

4. Mornings and backward leaning

Before getting up, on your back with knees bent: 10 slow belly breaths, 20 ankle pumps, 10 gentle knee rocks, 10 small pain-free pelvic tilts. Roll onto your side as one unit and push up with your arms.

Backward leaning. Your lowest spinal level no longer moves, so backward leaning must come from the level above and from your hips. Tight hip muscles and a tense, braced trunk make it hurt more.

- Avoid it during the first hour after getting up.
- Practise later in the day first, after a warm shower or a walk.
- Breathe out and relax your stomach before you lean; bracing increases pain.
- Start with a tiny range with hands on hips, 5 times, and increase slowly over weeks. This gradual exposure is a core part of cognitive functional therapy.
- Ask your physiotherapist about gentle hip-front stretches, which can take load off the back.

5. Sleep

- **Position:** side lying with a long pillow between knees and ankles, a small rolled towel under your waist and your neck level. Alternative: on your back with a pillow under your knees.
- **Fixed wake-up time** every day, including weekends. This is the strongest single sleep tool.
- **Bed is for sleep.** If awake more than 20 minutes, get up, sit somewhere dim and calm, return when sleepy.
- **Night-waking plan:** roll as one unit to a new position, slow breathing, heat. Note the time for your pain doctor.
- **Daylight in the morning, no caffeine after noon, short or no daytime naps.**
- **Ask for a sleep assessment.** Opioids can disturb deep sleep and cause breathing pauses at night. Snoring, morning headaches or daytime sleepiness are reasons to ask about a sleep study.
- Ask your general practitioner about cognitive behavioural therapy for insomnia, available as courses and online programmes.

6. Mind, mood and feeling stuck

Years of pain often bring low mood, worry, poor patience and a lost sense of who you are. This is common, it is not weakness, and treating it improves pain and function.

- **See a psychologist with pain experience.** Ask your general practitioner about a referral. Acceptance and commitment therapy, pain reprocessing therapy and cognitive behavioural therapy all have good evidence.
- **Live by your values now, not after the pain is gone.** Choose 1 to 3 things that matter (family, friends, work, a hobby) and set one small weekly goal for each.
- **Pace your days.** Spread activity evenly. Avoid doing a lot on good days and crashing afterwards.
- **Find others in the same situation.** Ask your municipality about the free course "Lær at tackle kroniske smerter", and look at Foreningen af Kroniske Smertepatienter for peer support.
- **Work.** If pain affects your job, contact your jobcenter, union or employer early about a gradual return, a work trial or adjusted tasks. The rules are changing, so ask what applies to you now.
- **Multidisciplinary pain centre.** If you are not already followed by one, ask your general practitioner about referral to Tværfagligt Smertecenter at Herlev and Gentofte Hospital or Rigshospitalet. They combine doctors, physiotherapists and psychologists.
- **If you have dark thoughts or feel hopeless,** call Livslinien on 70 201 201. You do not need to be in acute crisis to call.

7. Medication review: questions for your pain doctor

Never stop or change medication on your own. Stopping morphine or amitriptyline suddenly can cause withdrawal and worse pain.

What the research says, to discuss with your doctor:

- A one-year trial (Journal of the American Medical Association 2018) found long-term opioids no better than non-opioid medicines for function in chronic back, hip or knee pain.
- A 2024 review of pain-medication tapering found that pain on average improved rather than worsened, though study quality was mostly low.
- Low-dose naltrexone has not been tested in back pain. A 2025 review of chronic pain trials found no clear overall benefit, and a Danish fibromyalgia trial (The Lancet Rheumatology 2023) found no benefit over placebo.
- Naltrexone blocks opioids such as Malfin, so the two cannot simply be combined.

Questions:

- Is Malfin still helping my function, and could it be increasing my pain sensitivity?
- Is my night waking caused by medication wearing off?
- Could amitriptyline explain morning stiffness or grogginess? Do my medicines together affect my breathing at night or my heart rhythm?
- Is low-dose naltrexone suitable for me? What would coming off Malfin first involve, and how would it be supervised?
- Are there non-drug treatments I have not tried, such as a multidisciplinary pain programme?
- How do I prevent constipation, and is alcohol safe with my medicines?

8. Questions for your physiotherapist and surgeon

Physiotherapist:

- Can you deliver cognitive functional therapy, or refer me to someone who can?
- How far may I lean backward, and how do we build it up?
- When can I start exercises 9 and 12 in the hard program?
- Can I get warm-water training?

Surgeon:

- Does a computed tomography scan confirm the fusion is solid and the screws are secure?
- Could my pain come from the sacroiliac joints or the level above the fusion?
- Are there any movements or loads I should still avoid?

9. Other things that help

- **No smoking or nicotine:** it slows bone healing and worsens pain.
- **Sitting breaks** every 30 to 45 minutes.
- **Heat** for comfort before exercise and during flare-ups.
- **Regular meals with enough protein** to support muscle building.
- **Keep a short diary:** it makes appointments more useful.

Your first seven days

1. Book your physiotherapist and bring this guide.
2. Book your general practitioner: ask about a psychologist, the pain centre and inflammatory back pain tests.
3. Find your walking baseline and start.
4. Set a fixed wake-up time.
5. Write down three things that matter to you and one small goal for each.
6. Contact one person or group, so you are not doing this alone.
7. Start your weekly diary.

Warning signs: seek urgent care

Call 112, or the medical helpline 1813, immediately if you have:

- Numbness in the groin or buttocks, or loss of bladder or bowel control.
- New or worsening weakness or numbness in your legs.
- Fever with back pain, or a red, swollen or leaking scar.
- Sudden severe pain unlike your usual pain.
- Excessive drowsiness, confusion or slow breathing.

Key sources

- Kent and colleagues. Cognitive functional therapy (RESTORE). The Lancet, 2023; three-year follow-up, The Lancet Rheumatology, 2025.
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- Ashar and colleagues. Pain reprocessing therapy. Journal of the American Medical Association Psychiatry, 2022; five-year follow-up reported 2026.
- Rehabilitation to improve outcomes of lumbar fusion surgery: systematic review. European Spine Journal, 2022.
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- Manzetti and colleagues. Sacroiliac joint pain after lumbosacral fixation: systematic review. Clinical Spine Surgery, 2022.
- Spinal cord stimulation for persistent spinal pain syndrome type 2: systematic review, 2025.